Spontaneously Squeezing Superradiant Laser

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CLEO

Laser Science to Photonic Applications

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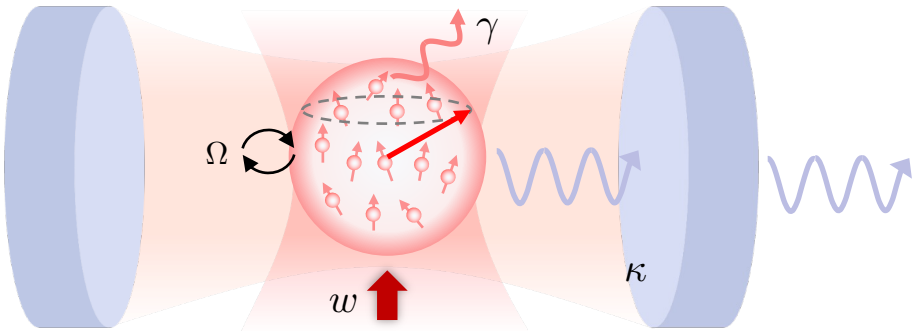
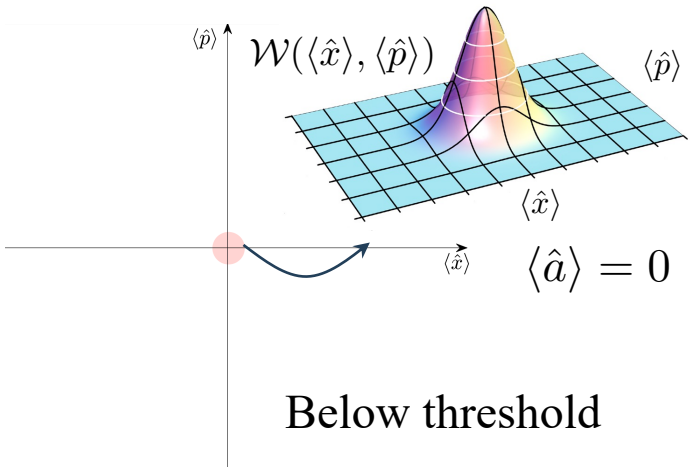
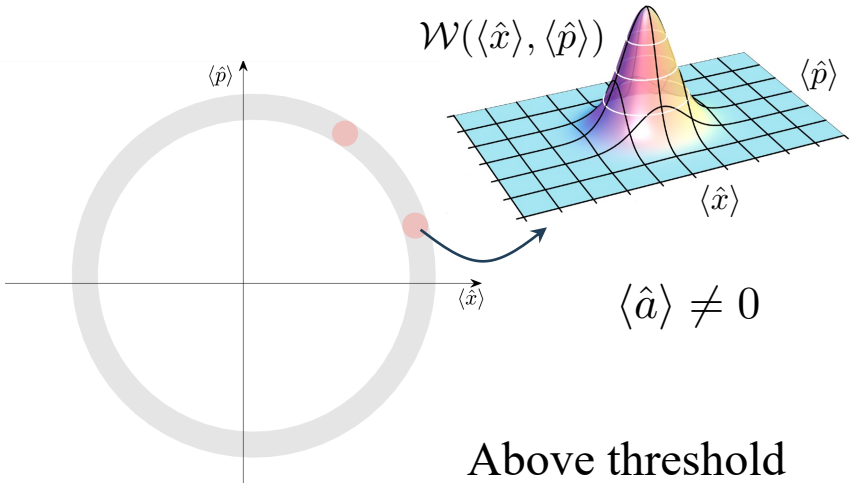
Superradiant Laser

- Coherent light sources
- Good coherence either from the collective atoms or from the cavity
- U(1) symmetry spontaneously broken above the threshold

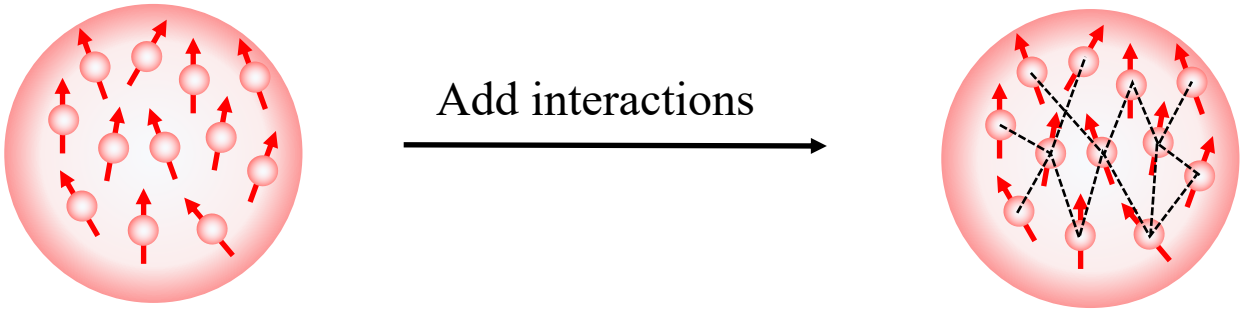






Carmichael, Howard J. Statistical methods in quantum optics. Springer Science & Business Media, 2013.

Atom-atom Interaction



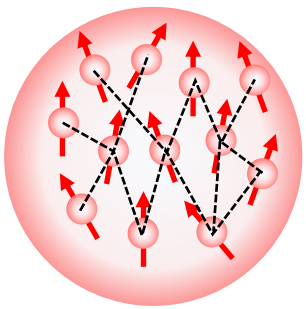
- Interaction is common: electric-dipole, spin-dipole, van der Walls, …
- Interacation brings correlations: atom-atom, atom-photon, photon-photon, …
- Interaction provide opportunities to generate quantum light
- …

Hutson, Ross B., et al. Science 383.6681, 384-387 (2024), …
Eckner, William J., et al. Nature 621, 734–739 (2023), ...

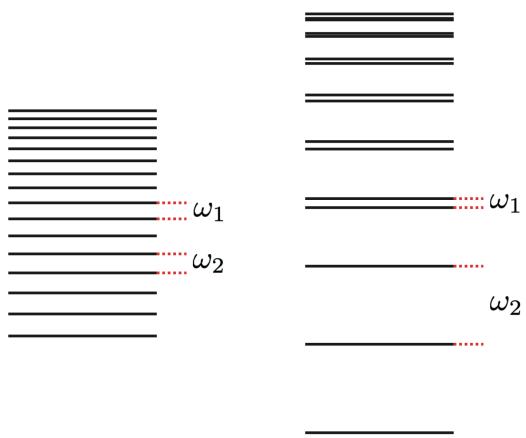
Interaction effect

➢ Lipkin-Meshkov-Glick(LMG) interaction

$$\hat{H}_{\mathrm{LMG}} = \frac{\lambda}{N}\sum_{i<j}\left(\hat{\sigma}_i^x\hat{\sigma}_j^x + \hat{\sigma}_i^y\hat{\sigma}_j^y\right) \propto -\frac{2\lambda}{N}\left(\hat{J}^z\right)^2$$



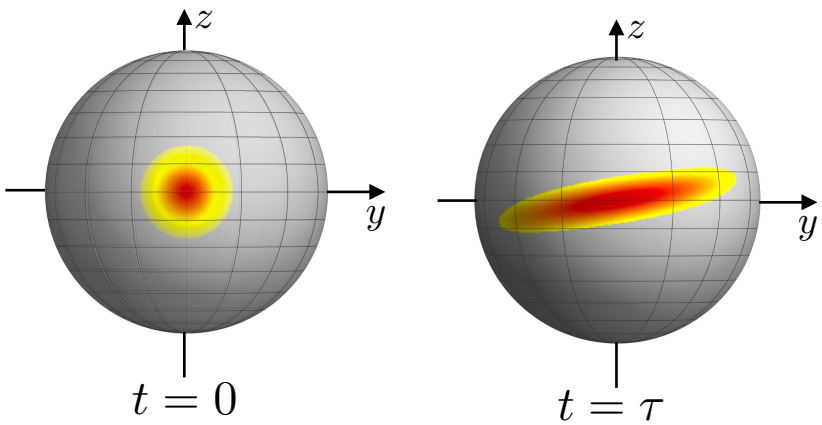
- Inhomogenous energy spectrum



Weak-interacting case

Strong-interacting case

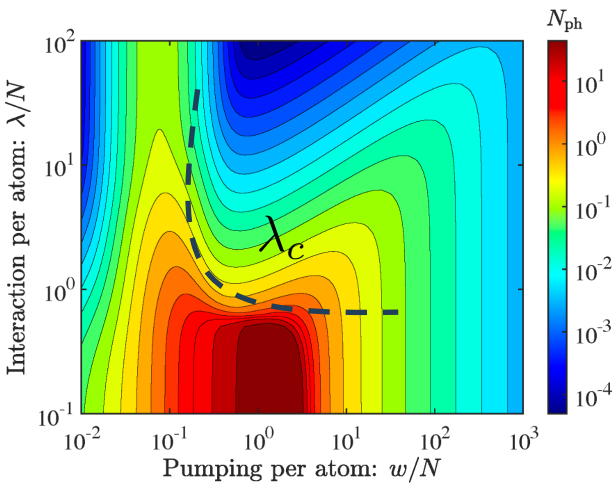
- Squeezing atoms

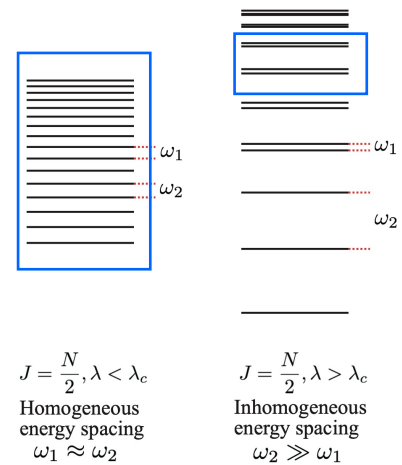


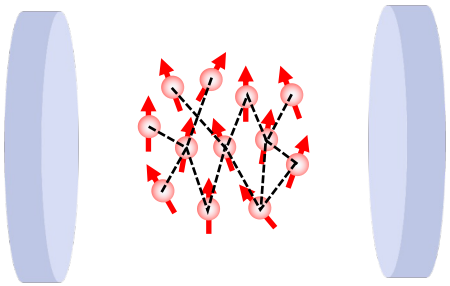
M. Kitagawa, M. Ueda, Phys. Rev. A 47, 5138 (1993), …

Blockade Lasing

Exact Numerical Simulation of N=15



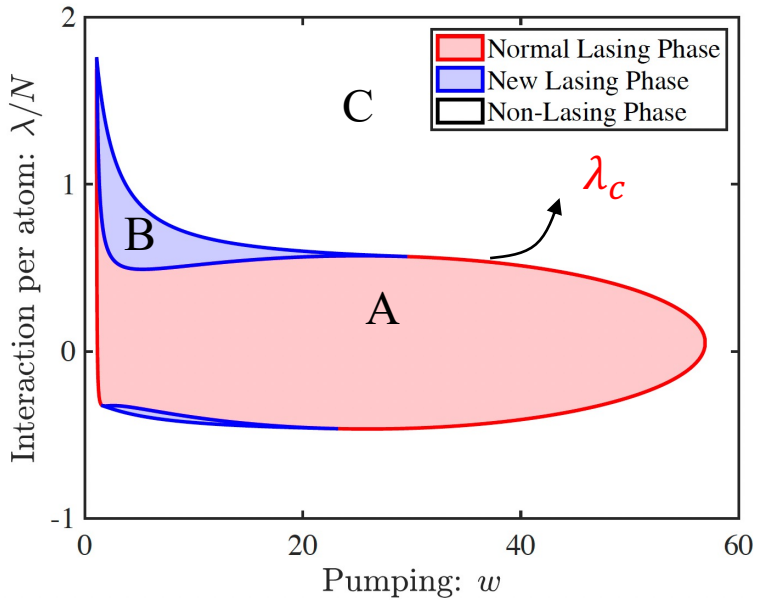




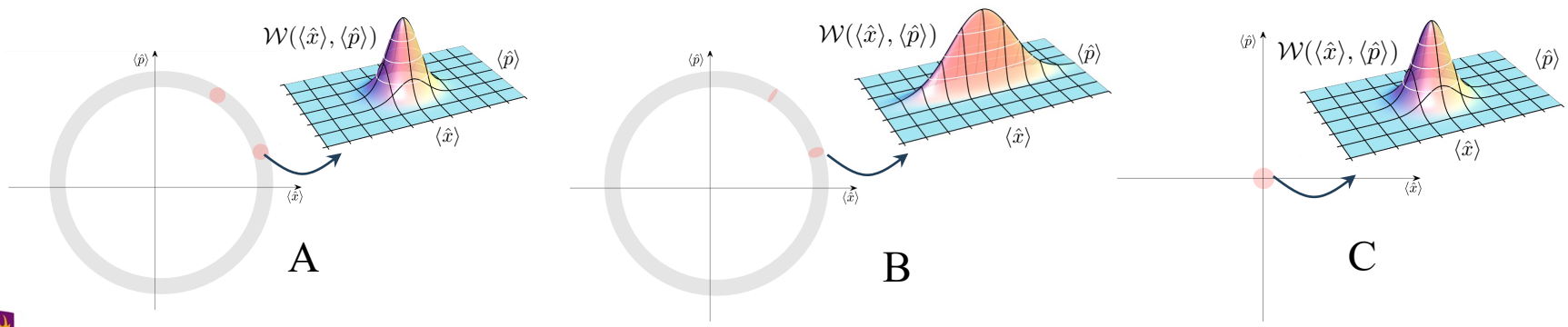
- Photon burst indicate a lasing and a threshold.
- In the weak-interacting case, all atoms transition are resonance with the cavity, which is normal lasing.
- In the strong-interacting case, small part of atoms transition are resonance with the cavity, which blockade lasing.
- Expect squeezing of atoms and cavity field in the strong-interacting case.

Urban, E., et al. Nature Physics 5, 110 (2009), …
B. A. Chase and J. M. Geremia, Phys. Rev. A 78, 052101 (2008), …

Phase diagram

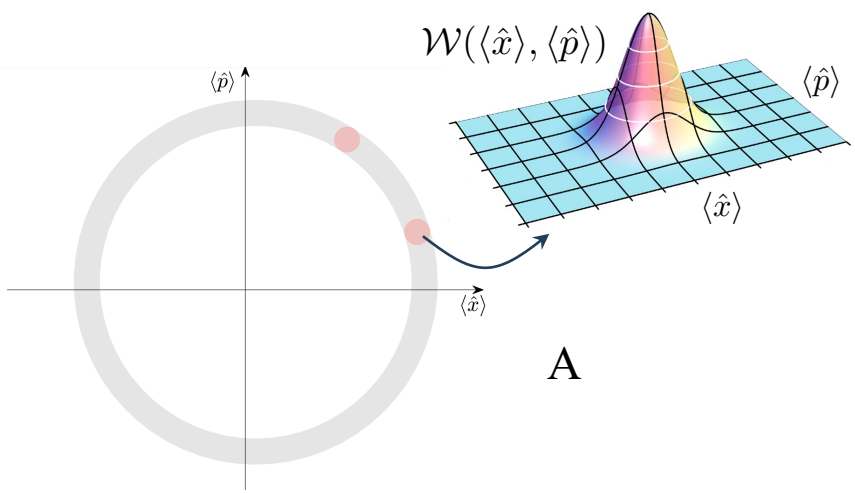


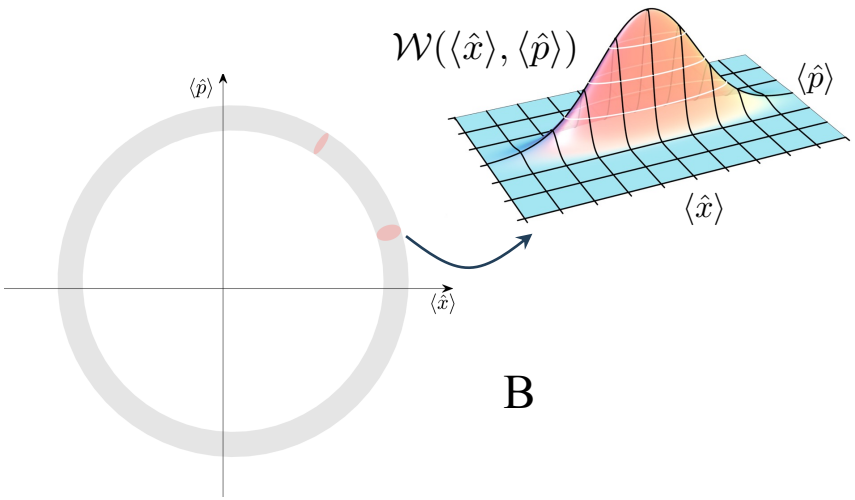
- Weak interacting phase A: normal lasing phase.
- Moderate interacting phase B: new lasing phase.
- Strong interacting phase C: non-lasing phase, which gives λ_c.



Spontaneous Squeezing Lasing

- The new lasing phase is a squeezed lasing phase.





The phase fluctuation is irrelevant to the amplitude flucutaitons

$$\phi \not\leftrightarrow \delta\hat{J}^z \leftrightarrow \delta\hat{a}^\dagger\delta\hat{a}$$

Normal Lasing: **Coherent state**

The phase fluctuation is correlated to the amplitude flucutaitons:

$$\phi \xleftrightarrow{\lambda} \delta\hat{J}^z \leftrightarrow \delta\hat{a}^\dagger\delta\hat{a}$$

Squeezed Lasing: **Squeezed state**

Squeezing Parameter

- The squeezing paramter

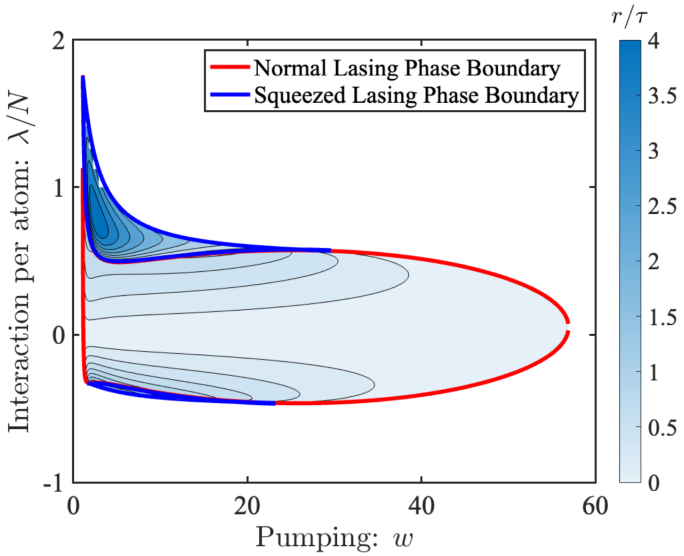
$$r = \xi\tau$$

$$\xi \propto \left|\frac{\lambda}{N}\right| (\mathscr{P} - \mathscr{P}_c)$$

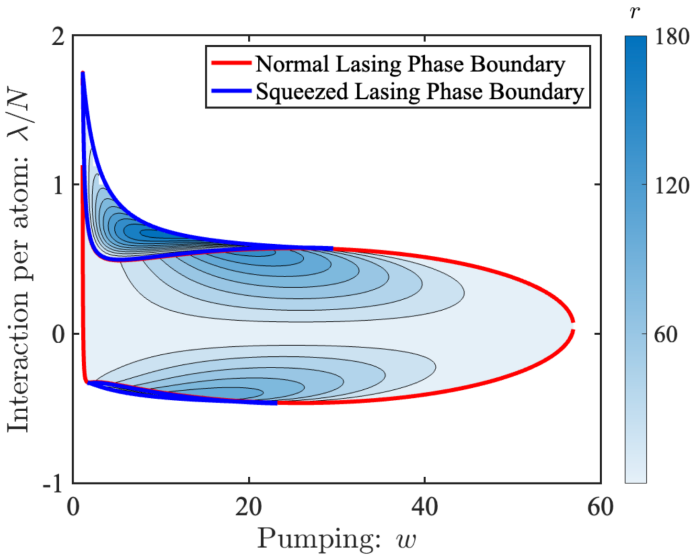
Squeezing efficiency

$$\tau$$

Squeezing time (i.e. phase correlation time)



Squeezing efficiency



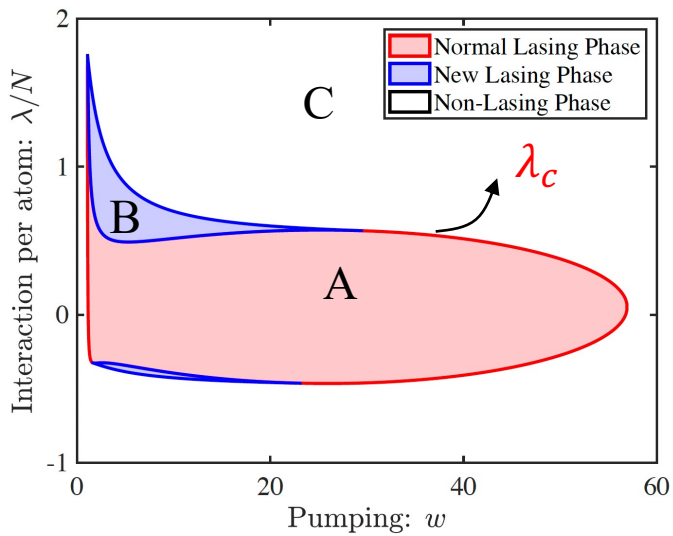
Squeezing parameter

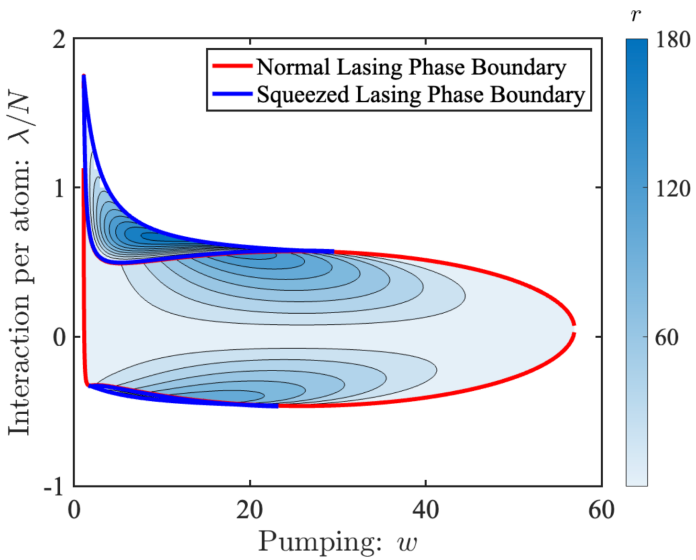
Experiment Predication

- Cooperative frequency shift $$\omega = \frac{(w+\gamma)\,\omega_c + \kappa\left(\omega_z - \boxed{\frac{4\lambda}{N}\left\langle \hat{J}^z \right\rangle}\right)}{w+\gamma+\kappa}$$
- Cooperative phase shift $$\Delta_{\mathrm{ps}} = \arctan \frac{2\left(\omega_c - \omega_z + \boxed{\frac{4\lambda}{N}\left\langle \hat{J}^z \right\rangle}\right)}{w+\gamma+\kappa}$$
- Steady squeezed phase $$r = \frac{\mathscr{P}_c\sqrt{\mathscr{P}_c}\left|\frac{\lambda}{N}\right|\kappa\,(w+\gamma)}{\Omega^2\,(w+\gamma+\kappa)}\,\frac{\mathscr{P}-\mathscr{P}_c}{\mathscr{P}}\,\kappa\tau$$
- First-order phase transition $$\frac{dr}{d\tau}$$ (finite size effect)
- …

Summary & outlook

- Study the collective effects of a many-body system, and discover a new squeezed lasing phase.
- Provide us a new tool to generate quantum light via manipulating atoms.
- Experiment verifcation, application of this quantum light, employ higher-order correlations in atoms to produce highly non-classical light, …





Da-Wu Xiao, et al. (In preparation)